

ApexLink Supply Chain Ltd

Data Analytics Revision Project

Company Overview

ApexLink Supply Chain Ltd is a Nigerian supply chain and logistics company specializing in the procurement, warehousing, and distribution of consumer goods across the country. The company partners with manufacturers and suppliers to deliver products efficiently to wholesalers, retailers, supermarkets, and corporate clients.

Founded in 2018, ApexLink has expanded its operations with multiple warehouses strategically located across Nigeria to ensure timely delivery and excellent customer service.

The company's primary color is Royal blue, secondary color is Orange, accent color is White and lastly, neutral color is Dark gray

Company Motto

"Delivering Excellence, Connecting Markets."

Vision Statement

To become Nigeria's most trusted and innovative supply chain company by delivering efficient, technology-driven logistics solutions that connect businesses and communities nationwide.

Mission Statement

To provide reliable, cost-effective, and customer-focused supply chain services through operational excellence, strategic partnerships, innovation, and continuous improvement while creating value for our customers, employees, and stakeholders.

As a Data Analyst, your task is to clean the data, analyse it, and present insights to management.

PART A – Microsoft Excel

Task 1 – Data Cleaning

Perform the following:

1. Make a backup copy of the original dataset.
2. Remove duplicate records.
3. Remove leading and trailing spaces.
4. Standardize text formatting using Proper Case where appropriate.
5. Correct inconsistent warehouse names

6. Correct inconsistent supplier names.
7. Correct inconsistent product names.
8. Correct inconsistent customer region names.
9. Standardize order status values.
10. Convert all dates to a single date format.
11. Convert Quantity, Unit Price, and Delivery Days to numeric values.
12. Remove or correct negative values where necessary.
13. Handle missing values appropriately.
14. Identify and treat outliers.
15. Ensure each Order_ID is unique after cleaning.

Task 2 – Create New Columns

Create the following calculated columns:

- **Total Sales**
- **Delivery Category**
 - 0–3 Days = Fast
 - 4–7 Days = Normal
 - Above 7 Days = Late
- **Month**
- **Quarter**
- **Year**

PART B – Excel Analysis

Using PivotTables and PivotCharts, answer the following:

1. Total Sales by Supplier.
2. Total Sales by Warehouse.
3. Total Sales by Customer Region.
4. Top 10 Products by Sales.

5. Order Status Distribution.
6. Monthly Sales Trend.
7. Average Delivery Days by Warehouse.
8. Average Delivery Days by Supplier.
9. Total Quantity Sold by Product.
10. Number of Orders by Region.

Create a simple Excel dashboard containing:

- KPI Cards (Total Sales, Total Orders, Average Delivery Days)
- At least four charts
- Slicers for Supplier, Warehouse, Region, and Order Status

PART C – MySQL

Database Creation

1. Create a database named ApexLink.
2. Create a table matching the dataset structure.
3. Import the cleaned dataset into MySQL.

SQL Queries

Write SQL queries to answer the following:

Basic

1. Display all records.
2. Display only Supplier and Product columns.
3. Display the first 20 records.
4. Show all distinct suppliers.
5. Show all distinct warehouses.

Filtering

6. Find orders with Quantity greater than 50.
7. Find products with Unit Price between 5,000 and 20,000.

8. Find all Delivered orders.
9. Find all Pending orders.
10. Find all Cancelled orders.
11. Find all orders from Lagos warehouse.
12. Find suppliers whose names begin with "D".
13. Find suppliers containing "Nest".
14. Find products ending with "Rice".
15. Find orders where Delivery Days is greater than 7.

Aggregate Functions

16. Total Sales.
17. Average Unit Price.
18. Average Delivery Days.
19. Maximum Unit Price.
20. Minimum Unit Price.
21. Total Quantity Sold.

GROUP BY

22. Total Sales by Supplier.
23. Total Sales by Warehouse.
24. Total Sales by Customer Region.
25. Number of Orders by Supplier.
26. Number of Orders by Warehouse.
27. Number of Orders by Status.

HAVING

28. Suppliers with more than 20 orders.
29. Warehouses with average delivery days greater than 5.
30. Products with total quantity sold greater than 100.

CASE Statement

31. Create a column named **Delivery Performance**:

- ≤ 3 days $\rightarrow$ Fast
- 4–7 days $\rightarrow$ Normal
- 7 days $\rightarrow$ Late

Views

32. Create a view for Delivered Orders.

33. Create a view for Late Deliveries.

34. Create a view showing Total Sales by Supplier.

PART D – Power BI

Import the cleaned Excel dataset into Power BI.

Data Preparation

1. Ensure all columns have the correct data type.
2. Rename fields where necessary.
3. Remove any remaining errors.
4. Create a Date hierarchy.

DAX Measures

Create the following measures:

1. Total Sales
2. Total Orders
3. Average Delivery Days
4. Average Unit Price
5. Total Quantity Sold
6. Late Delivery Percentage
7. Delivered Orders
8. Cancelled Orders
9. Pending Orders

Dashboard

Create a dashboard with the following visuals:

1. Card – Total Sales
2. Card – Total Orders
3. Card – Average Delivery Days
4. Clustered Bar Chart – Sales by Supplier
5. Column Chart – Sales by Warehouse
6. Pie Chart – Order Status Distribution
7. Line Chart – Monthly Sales Trend
8. Treemap – Sales by Product
9. Matrix – Supplier vs Warehouse
10. Table – Top 10 Products by Sales

Slicers

Add slicers for:

- Supplier
- Warehouse
- Customer Region
- Order Status
- Month

PART E – Business Questions

Using your analysis, answer the following:

1. Which supplier generated the highest sales?
2. Which warehouse handled the highest sales volume?
3. Which product generated the highest revenue?
4. Which customer region generated the most sales?
5. Which warehouse had the highest average delivery time?
6. Which supplier had the most late deliveries?

7. Which products should be restocked?
8. Which products should be discontinued?
9. What causes delivery delays?
10. How can operational costs be reduced while improving customer satisfaction?
11. Which order status occurred most frequently?
12. Which month recorded the highest sales?
13. What are the top five best-performing products?
14. Based on your findings, provide **five recommendations** that management can implement to improve operational efficiency.